



MEDIAN OF TWO SORTED ARRAYS 64

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Given two sorted arrays, merge them and find the median of the combined sorted array.

Input Format

First line contains two integers n and m . Second line contains n sorted integers. Third line contains m sorted integers.

Constraints

$1 \leq n, m \leq 10^5$ $-10^6 \leq \text{array elements} \leq 10^6$

Output Format

Print the median of the merged array.

If the total number of elements is odd, print the middle element. If even, print the average of the two middle elements with one decimal place.

Sample Input 0

```
2 2
1 2
3 4
```

Sample Output 0

```
2.5
```

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Language

C



```
1  #include <stdio.h>
2  #include <string.h>
3  #include <math.h>
4  #include <stdlib.h>
5  void mergesort(int arr1[],int m,int arr2[],int n,int result[]){
6      int i=0,j=0,k=0;
7      while(i<m && j<n){
8          if(arr1[i]<arr2[j]){
9              result[k++]=arr1[i++];
10         }
11         else{
12             result[k++]=arr2[j++];
13         }
14     }
15     while(i<m){
16         result[k++]=arr1[i++];
17     }
18     while(j<n){
19         result[k++]=arr2[j++];
20     }
21     if(k%2==0){
22         float key = ((float)result[k/2]+(float)result[k/2-1])/2.0;
23         printf("%.1f",key);
24     }
25     else{
26         printf("%.1f",(float)result[k/2]);
27     }
28 }
```



```
29  int main() {
30      int m,n;
31      int arr1[100000],arr2[100000];
32      int result[200000];
33      scanf("%d",&m);
34      scanf("%d",&n);
35      for(int i=0;i<m;i++){
36          scanf("%d",&arr1[i]);
37      }
38      for(int i=0;i<n;i++){
39          scanf("%d",&arr2[i]);
40      }
41      mergesort(arr1,m,arr2,n,result);
42  }
43
```

Line: 43 Col: 1

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